

Representation choices for somatic mutation catalogues strongly influence prediction performance across mechanistic and clinical endpoints

Aaron Ge^{1,2}, Jeya Balaji Balasubramanian³, Jonas De Almeida³, Bradley A. Maron^{1,2*}

¹ University of Maryland-Institute of Health Computing, University of Maryland School of Medicine, Baltimore, MD, USA

² University of Maryland School of Medicine, Baltimore, MD, USA

³ Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, Bethesda, MD, USA

Correspondence: Aaron Ge

6116 Executive Blvd, Bethesda, MD 20282, USA

age1@som.umaryland.edu

Abstract

Background. Cancer sequencing produces variable-length catalogues of somatic mutation events. These catalogues can be summarized as mutation burdens, mutational spectra, event-level annotation features, or learned neural event bags, but it remains unclear how much prediction performance depends on representation choice rather than model family alone.

Methods. We regenerated a strict no-leakage benchmark spanning five main endpoints: Kucab DNA damage class ($n = 259$), TCGA-BRCA HRD score ($n = 772$), HRD-high status at threshold 33 ($n = 772$), TCGA cancer type among the ten most common classes ($n = 5,462$), and TCGA CDR overall survival ($n = 9,986$). Representations included mutational burden, SBS/DBS/ID mutational spectra, event-level MAF-stack features, and combined Signatures + MAF stack features. Tabular learners used five outer folds, an inner train/validation split for hyperparameter selection, final refitting on the full outer-training fold, and pooled out-of-fold metrics. Survival was modeled with Cox proportional hazards and Harrell's C-index. We also added a MuAt-compatible event-bag comparator on the same manuscript endpoints and outer folds where possible.

Results. Mutational signatures improved over burden for XGBoost on Kucab damage class, HRD score, HRD33 high/low status, and cancer-type prediction. The strongest tabular performance generally came from Signatures + MAF stack with XGBoost: HRD score reached 0.749 Spearman r , HRD33 high/low reached 0.886 AUROC, and cancer type reached 0.969 macro-AUROC. Overall survival was best predicted by mutational signatures under Cox PH (C-index 0.607), while richer event-level MAF features did not improve this survival endpoint. The MuAt-compatible comparator reached 0.928 macro-AUROC for cancer type on the directly comparable TCGA-WES top-10 task, below the tuned Signatures + MAF stack XGBoost baseline.

Conclusions. Representation choice is a first-order determinant of prediction performance for somatic-mutation benchmarks. Mutational signatures remain a strong and efficient baseline, event-level MAF features add endpoint-specific biology, and combining spectra with event-level annotations is the most consistent practical tabular default. A local MuAt-compatible reimplementaion does not automatically exceed a tuned tabular baseline under a directly comparable TCGA-WES evaluation.

Trial registration. Not applicable.

Keywords: Somatic mutations; Mutational signatures; Mutation annotation format; Representation learning; Homologous recombination deficiency; Tumour type; Survival analysis; Cox proportional hazards; MuAt; XGBoost

Background

Somatic mutation catalogues integrate the activity of endogenous and exogenous mutational processes, DNA repair defects, and clonal selection. Large pan-cancer resources have catalogued SBS, DBS, and indel signatures across whole genomes and exomes, while controlled mutagen-exposure experiments have shown that many environmental agents leave reproducible sequence-context spectra [1,2]. These observations make mutation catalogues a natural substrate for mechanistic attribution and clinical prediction.

A practical prediction pipeline must decide how to transform a variable-length set of mutation events into model inputs. Common tabular choices include mutation burden, mutational spectra, and aggregated MAF-derived gene, locus, consequence, pathway, and variant-allele-fraction features. More recent neural approaches, including Mutation-Attention (MuAt), instead operate on mutation-level events and learn attention-based tumour representations [10].

The manuscript question is therefore not whether mutation catalogues contain predictive information, but how strongly the chosen representation shapes performance when evaluation is held fixed. We address critiques of exome sparsity, insufficient tuning, survival modeling, and deep-learning comparability by regenerating the benchmark under a strict no-leakage, nested out-of-fold protocol with Cox survival endpoints and a directly comparable MuAt-compatible event-bag model.

Figure 1. Mutation catalogues to benchmarked representations

Primary path: mutation events are converted into tabular feature families and evaluated with paired 5-fold out-of-fold endpoint metrics; the MuAt-compatible event-bag model is shown as a separate direct comparator, not as a tabular learner.

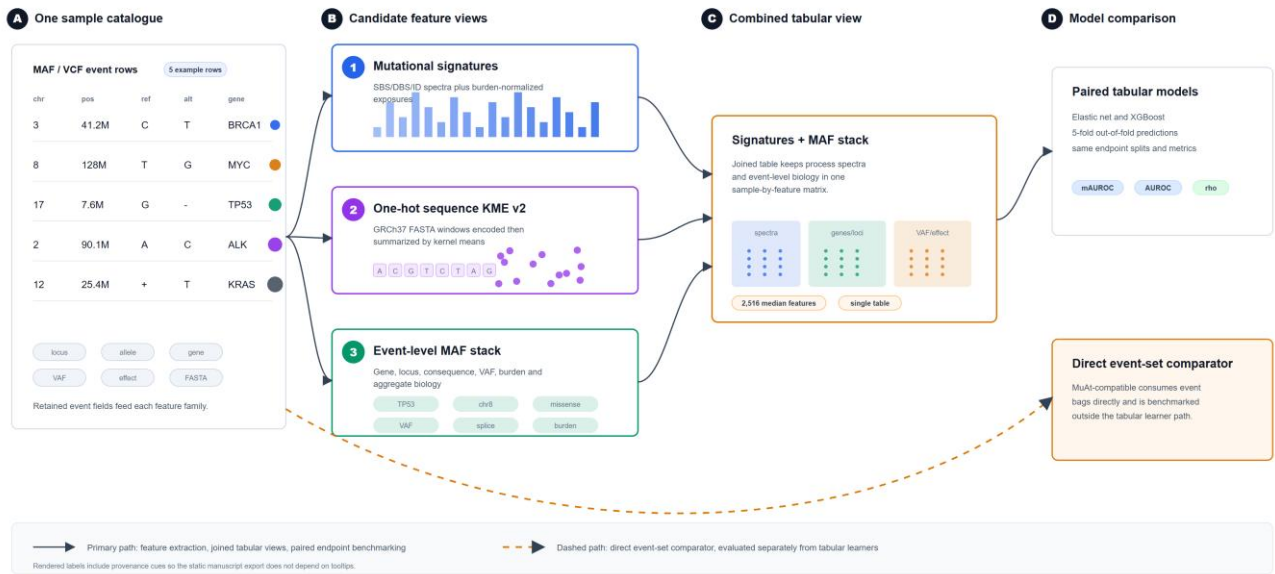


Figure 1. Conceptual overview of mutation-catalogue representations.

Each tumour is represented as a catalogue of somatic mutation events that can be transformed into compact burden summaries, mutational spectra, event-level MAF-stack aggregates, combined tabular representations, or MuAt-compatible mutation bags. The benchmark evaluates these choices under a strict pooled out-of-fold design.

Methods

Study design and endpoint registry

All analyses were regenerated inside the project repository using an endpoint registry and strict completion gate. The main panel contains five endpoints chosen to span mechanism, DNA repair deficiency, tumour identity, and clinical outcome. Binary overall-survival classification was removed from the main benchmark and replaced by Cox proportional hazards modeling of TCGA CDR overall survival.

The five primary endpoints were Kucab DNA damage class; continuous HRD score; HRD-high versus HRD-low at threshold 33; TCGA cancer type among the ten most common classes; and TCGA CDR overall survival. Survival endpoints required positive follow-up time, at least 200 eligible samples, and at least 25 events.

Table 1. Datasets, endpoints, and evaluation design.

Endpoint	Cohort/source	N	Task	Primary metric
Kucab damage class	Kucab mutagen-treated clone mutation catalogues	259	Grouped multiclass Multiclass classification	macro-AUROC
HRD score	TCGA-BRCA HRD labels with MC3 mutation features	772	Regression	Spearman r
HRD33 high/low	TCGA-BRCA HRD labels with MC3 mutation features	772	Binary classification	AUROC
Cancer type (top 10)	TCGA MC3 clinical endpoint panel	5,462	Multiclass classification	macro-AUROC
Overall survival	TCGA MC3 clinical endpoint panel	9,986	Survival	Harrell C-index

Note. Label definitions are provided in the endpoint registry; survival is modeled as a time-to-event endpoint with censoring rather than as binary event status.

Representations

Mutational burden features provide the minimal count-based baseline. Mutational signatures summarize SBS, DBS, and indel spectra. Event-level MAF-stack features aggregate mutation-level gene, locus, consequence, pathway, variant-allele-fraction, and topographic information within training-safe feature dictionaries. Signatures + MAF stack concatenates mutational spectra with event-level annotation features.

For the neural comparator, a MuAt-compatible preprocessing layer turns each mutation into three token streams: a mutation motif token, a 1-Mb chromosome-position-bin token, and an annotation token derived from available MAF fields such as gene, exon/consequence, and strand. Tokens are mapped with fixed hash dictionaries rather than cohort-learned vocabularies, preventing held-out test samples from influencing the available token set.

Table 3. Representation summary and dimensionality.

Representation	Input signal	Dimensionality	Evaluated models
Mutational burden	Total mutation burden and compact burden summaries	1-3	Elastic net; XGBoost; Cox PH
Mutational signatures	SBS/DBS/ID mutational spectra	182-261	Elastic net; XGBoost; Cox PH
Event-level MAF stack	Gene, locus, consequence, VAF, and event-level MAF aggregates	1,927-2,980	Elastic net; XGBoost; Cox PH
Signatures + MAF stack	Concatenated mutational spectra and event-level MAF aggregates	2,109-3,241	Elastic net; XGBoost; Cox PH
MuAt-compatible	Per-mutation MC3 event bags with motif, position-bin, and genic/exonic/strand tokens	event bag	MuAt-compatible

Note. Feature dimensionality is reported for tabular matrices; MuAt-compatible uses padded mutation bags and learned embeddings rather than a fixed tabular feature count.

Nested out-of-fold evaluation

Each endpoint used five outer folds. For every outer fold, the 80% outer-training partition was split into inner-training and inner-validation partitions. Candidate models were trained only on the inner-training set, selected only on the inner-validation set, refit once on the full 80% outer-training set using the selected hyperparameters, and predicted once on the 20% held-out outer fold. Primary metrics were computed after pooling all out-of-fold predictions over the full cohort.

Classification used stratified splits when feasible; survival used event-stratified splits; regression used shuffled K-fold splits; and Kucab analyses used grouped splitting to prevent treatment-agent leakage across folds.

Logistic elastic-net models searched C values of 0.001, 0.01, 0.1, 1, and 10 with l1_ratio values of 0, 0.25, 0.5, 0.75, and 1. Regression elastic net and Cox PH models searched alpha or penalizer values from 1e-4 to 10 with the same l1_ratio grid. XGBoost used seeded candidate search with inner-validation early stopping; final refits used exactly the selected number of boosting rounds.

Primary metrics were Spearman r for continuous endpoints, AUROC for binary endpoints, macro-AUROC for multiclass endpoints, and Harrell C-index for survival endpoints. Multiclass Kucab robustness was further summarized with micro-AUROC, balanced accuracy, macro-F1, and Cohen's kappa. Statistical comparisons used paired tests over pooled out-of-fold predictions with Benjamini-Hochberg FDR correction.

MuAt-compatible comparator

The MuAt-compatible comparator follows the central structure reported for MuAt: separate motif, position, and annotation embeddings; concatenation of modality embeddings; Q/K/V self-attention; residual/normalization layers; a fully connected block; average pooling; and a 24-dimensional tumour feature layer before prediction. We label this model MuAt-compatible rather than MuAt because official MuAt package checkpoints were not used for the manuscript comparison.

The primary manuscript comparison evaluates this neural event-bag model on the same TCGA-WES endpoints as the tabular benchmark. The TCGA 20-type task from the original MuAt paper was retained only as a supplemental fidelity artifact, because it is not directly comparable to the paper's main top-10 cancer-type endpoint and does not share the same fold structure or endpoint definition.

Results

Mutational signatures improve over burden, but not uniformly

In the strict main panel, XGBoost models using mutational signatures outperformed burden on Kucab damage class (0.636 vs 0.505 macro-AUROC), HRD score (0.704 vs 0.596 Spearman r), HRD33 high/low status (0.845 vs 0.811 AUROC), and cancer-type prediction (0.939 vs 0.840 macro-AUROC). Elastic-net models showed the same qualitative gain for Kucab damage class and cancer type but not for every clinical endpoint, underscoring that signatures are strong but not universally dominant.

The survival result should be interpreted separately from the older binary-OS draft: overall survival is now a Cox PH endpoint evaluated by Harrell C-index. Under this survival analysis, mutational signatures achieved the strongest overall-survival C-index among the main tabular representations (0.607).

Figure 2. Burden and signature baselines

5-fold OOF scores with paired tests. Metrics: macro-AUROC, Spearman r, AUROC, Harrell C-index.

Endpoint labels show metric, task, and n. Delta intervals are paired 95% bootstrap CIs where available.

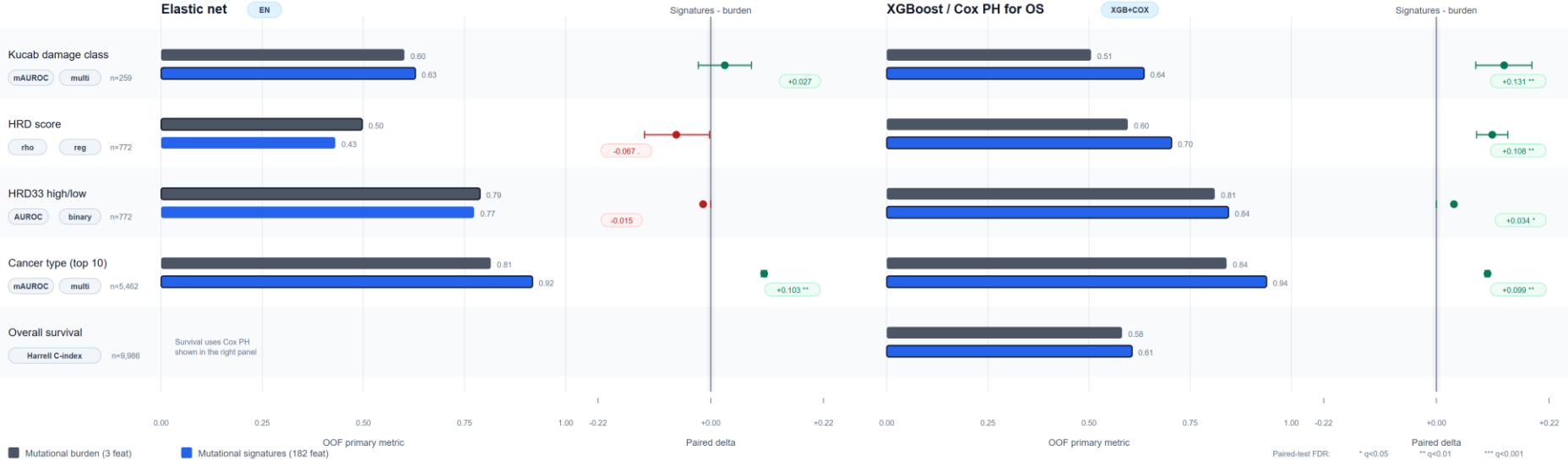


Figure 2. Signature baselines compared with mutational burden.

Nested five-fold out-of-fold performance is shown for burden and mutational signatures across the five main endpoints. Metrics are Spearman r for HRD score, AUROC for HRD33, macro-AUROC for multiclass endpoints, and Harrell C-index for survival.

Event-level MAF features add endpoint-specific biology

Event-level MAF-stack features were most useful for endpoints where gene, locus, consequence, or event-level biology carries direct label information. With XGBoost, MAF stack alone improved cancer-type prediction relative to signatures (0.950 vs 0.939 macro-AUROC), but it underperformed signatures for the mechanistic Kucab damage-class task (0.529 vs 0.636 macro-AUROC). This pattern supports the expected distinction between mutagen-mechanism endpoints, which are strongly sequence-context driven, and clinical or tumour-identity endpoints, which can benefit from broader annotation features.

The combined Signatures + MAF stack representation was the strongest practical tabular default. With XGBoost it reached 0.749 Spearman r for HRD score, 0.886 AUROC for HRD33 high/low, and 0.969 macro-AUROC for cancer type. It did not improve overall survival, where signatures alone remained best under Cox PH.

Figure 4. MAF-stack biology vs signatures

Bars show canonical 5-fold OOF performance. Right panels show signed pairwise deltas and FDR markers.

Contrast signs are candidate minus baseline. Intervals are shown for bootstrap tests; DeLong rows carry p/q markers without bootstrap CIs.

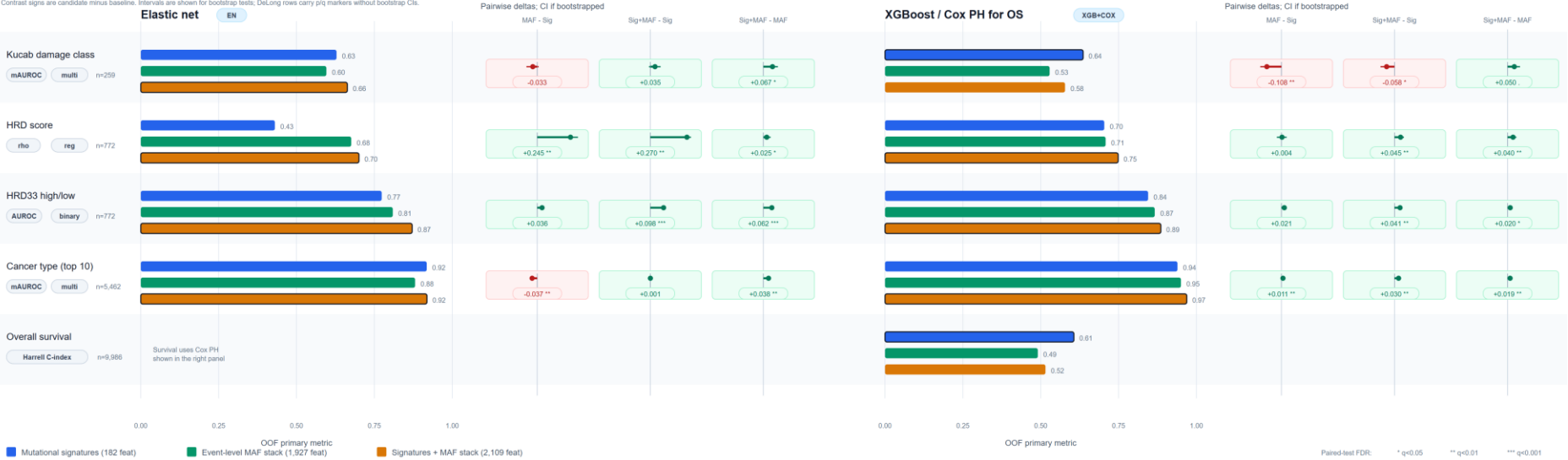


Figure 4. Event-level MAF-stack features and combined signature-plus-event representations.

Mutational signatures, event-level MAF-stack features, and their concatenation are compared for each endpoint and tabular model family. The combined representation is strongest for HRD score, HRD33 status, and cancer type under XGBoost.

MuAt-compatible event-bag modeling is directly comparable but not dominant

The MuAt-compatible comparator was added to avoid treating deep event-level models only as a conceptual future baseline. On cancer type among the same ten TCGA classes, it reached 0.928 macro-AUROC. This exceeds several simpler representations but remains below the tuned Signatures + MAF stack XGBoost model at 0.969. On HRD score, HRD33 high/low, Kucab damage class, and overall survival, the MuAt-compatible model also did not exceed the best tabular result.

These results do not imply that the official pretrained MuAt model is ineffective. They show that, in this manuscript's strict TCGA-WES benchmark and local reimplementation, a mutation-attention architecture does not automatically outperform a leakage-safe, tuned tabular representation.

Figure 3. MuAt-compatible event-bag comparator

Direct event-bag model, compared with the best tabular row and the Signatures + MAF default on pooled OOF metrics.



Figure 3. MuAt-compatible event-bag comparator.

The MuAt-compatible reimplemention is evaluated on the manuscript endpoints rather than as a reproduction of the original MuAt TCGA-20 benchmark. For TCGA endpoints, the comparator uses the same held-out folds as the canonical tabular benchmark where configured.

Table 2. Main-panel performance by endpoint and representation.

Endpoint / representation	Elastic net	XGBoost / Cox PH	MuAt-compatible
Kucab damage class (macro-AUROC; n=259)			
Mutational burden	0.601	0.505	
Mutational signatures	0.629	0.636	
Event-level MAF stack	0.596	0.529	
Signatures + MAF stack	0.663	0.578	
MuAt-compatible event bag			0.599
HRD score (Spearman r; n=772)			
Mutational burden	0.497	0.596	
Mutational signatures	0.431	0.704	
Event-level MAF stack	0.676	0.708	
Signatures + MAF stack	0.701	0.749	
MuAt-compatible event bag			0.492
HRD33 high/low (AUROC; n=772)			
Mutational burden	0.788	0.811	
Mutational signatures	0.774	0.845	
Event-level MAF stack	0.809	0.866	
Signatures + MAF stack	0.872	0.886	
MuAt-compatible event bag			0.781
Cancer type (top 10) (macro-AUROC; n=5,462)			
Mutational burden	0.815	0.840	
Mutational signatures	0.918	0.939	
Event-level MAF stack	0.881	0.950	
Signatures + MAF stack	0.919	0.969	
MuAt-compatible event bag			0.928
Overall survival (Harrell C-index; n=9,986)			
Mutational burden		0.582	
Mutational signatures		0.607	
Event-level MAF stack		0.491	
Signatures + MAF stack		0.515	
MuAt-compatible event bag			0.601

Note. Scores are pooled out-of-fold primary metrics. Elastic net and XGBoost columns apply to tabular representations; the Cox PH value is shown in the XGBoost/Cox PH column for survival. MuAt-compatible is shown as a separate event-bag comparator.

Cross-endpoint synthesis

No representation wins across every endpoint. Mutational signatures are a strong and efficient baseline; event-level MAF features capture biology that spectra can miss; and combined signature-plus-MAF features offer the best overall tabular default for HRD and tumour-identity endpoints. Survival remains difficult, and richer annotation stacks did not improve over signatures in the current Cox PH analysis.

The Kucab sparsity control downsampled WGS mutation inventories to budgets of 20, 41, 100, and 200 mutations per clone. These runs provide an exome-like stress test for the mechanistic endpoint and are reported in the supplementary tables. They make explicit that WGS-versus-WES mutation count differences can affect representation stability and should not be ignored when comparing mechanistic and clinical cohorts.

Figure 5. Cross-endpoint representation summary

Absolute canonical 5-fold OOF scores. MuAt-compatible is separated as a direct event-bag model, not nested under Elastic net or XGBoost.

Black outline marks the best displayed score within each endpoint. OS is evaluated with Cox PH; Elastic net cells are intentionally not applicable for survival.

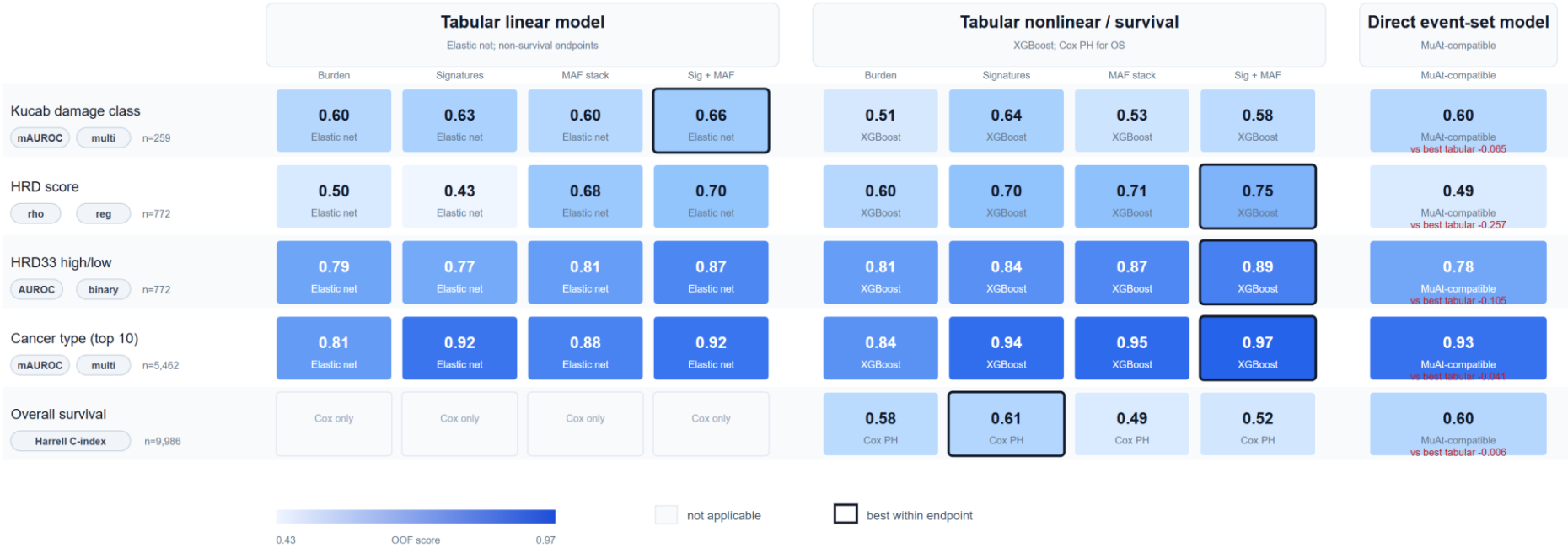


Figure 5. Cross-endpoint representation summary.

The summary heatmap reports the canonical main-panel scores. MuAt-compatible results are displayed as a separate event-bag comparator, and overall survival is reported as a Cox PH C-index rather than a binary event AUROC.

Discussion

This regenerated benchmark changes the interpretation of the original draft in three important ways. First, survival is now evaluated as time-to-event data with censoring rather than as binary event status. Second, high-dimensional tabular baselines are tuned within a nested out-of-fold framework, reducing the risk that poor performance reflects inadequate regularization. Third, the neural event-bag comparator is measured directly rather than invoked as an untested future alternative.

The results support a pragmatic view of mutation-catalogue modeling. For mechanistic mutagen attribution, mutational spectra remain highly competitive because the label is closely tied to sequence-context processes. For HRD and cancer-type endpoints, event-level annotations add useful biological structure, and the combined Signatures + MAF stack representation performs best with XGBoost. For survival, predictive signal from mutation catalogues alone remains modest and appears sensitive to representation and censoring-aware modeling assumptions.

The MuAt-compatible model's weaker performance relative to tuned tabular models is scientifically plausible. TCGA WES mutation bags are sparse, the model was trained locally rather than initialized from official MuAt checkpoints, and the manuscript endpoints include HRD and survival tasks that were not the original MuAt paper's primary tumour-typing benchmark. The result should therefore be read as a fair local comparator, not as a failure to reproduce the full MuAt PCAWG/TCGA/GEL study.

Limitations

The benchmark uses bundled MC3, CDR, HRD, and Kucab assets rather than downloading new GDC cohorts during the default run. GDC expansion utilities are available but disabled by default to keep the manuscript run offline and reproducible from the bundled assets.

The MuAt-compatible comparator implements the central mutation-attention architecture and token modalities but does not include official pretrained checkpoints, external WGS validation cohorts, structural-variant and mobile-element modalities unavailable in the bundled TCGA WES assets, or the full 150-epoch MuAt paper training regimen.

Some comparisons involving MuAt and simpler tabular representations share the same endpoint and out-of-fold pooling but are not all paired on identical fold manifests for every representation family. The strongest TCGA cancer-type comparison was aligned to the canonical tabular fold assignment, and this is the comparison emphasized for direct model interpretation.

Conclusions

Representation choice strongly influences somatic-mutation prediction performance. Mutational signatures are a robust baseline, event-level MAF features add complementary endpoint-specific signal, and their combination with XGBoost is the strongest practical default for several clinical and tumour-identity endpoints. Cox survival modeling and no-leakage feature generation materially strengthen the clinical interpretation of the benchmark. A MuAt-compatible event-bag comparator provides a direct neural-model comparison but does not supersede the tuned tabular baseline under the current TCGA-WES evaluation.

Abbreviations

Table 4. Key terminology and abbreviations.

Term	Definition	Used in
EN / XGB	Elastic-net score / XGBoost score in compact performance tables.	Table 2
AUROC	Area under the receiver operating characteristic curve for binary endpoints.	Tables 2 and S2
macro-AUROC	Class-balanced AUROC averaged across multiclass labels.	Tables 2 and S2
Spearman r	Spearman rank correlation for continuous endpoints.	Tables 2 and S2
HRD	Homologous recombination deficiency.	Endpoint labels
MAF	Mutation annotation format; here also shorthand for event-level mutation annotations.	Representations
SBS/DBS/ID	Single-base substitution, double-base substitution, and insertion/deletion mutation spectra.	Representations
KME	Kernel mean embedding used to summarize event-level mutation contexts.	Representations
UGA	Universal genomic atlas/channel geometry representation used in supplementary analyses.	Supplement
Mutational burden	Compact mutation-count baseline features.	Tables 2 and 3
Mutational signatures	Canonical mutation-spectrum representation.	Tables 2 and 3
One-hot sequence KME	FASTA-window sequence-context KME representation.	Tables 2 and 3
Event-level MAF stack	Gene, locus, consequence, VAF, and event-annotation feature stack.	Tables 2 and 3
Signatures + MAF stack	Combined spectra plus event-level MAF features.	Tables 2 and 3
XGBoost	Gradient-boosted tree model used as the nonlinear learner.	Model columns

Note. Abbreviations and shorthand used in the manuscript figures and tables.

Declarations

Ethics approval and consent to participate

Not applicable. The analyses use public, de-identified research datasets.

Consent for publication

Not applicable.

Availability of data and materials

All analyses were implemented in the GitHub-connected project repository. The default reproducibility run uses bundled MC3, TCGA CDR, TCGA-BRCA HRD, and Kucab assets, with optional GDC utilities disabled by default. Generated tables, figures, split manifests, and audit files are stored under the project results directory.

Competing interests

The authors declare that they have no competing interests.

Funding

The research reported here was supported by the AIM-AHEAD Coordinating Center at the University of North Texas Health Science Center at Fort Worth and, in part, by the National Institutes of Health Agreement No. 1OT2OD032581. The views and conclusions are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the NIH.

Authors' contributions

AG designed and implemented the benchmark, performed analyses, generated figures and tables, and drafted the manuscript. JBB and JDA contributed cancer-genomics and benchmarking expertise. BAM supervised the project and revised the manuscript. All authors reviewed and approved the manuscript.

Acknowledgements

The authors thank the teams responsible for TCGA, MC3, TCGA CDR, HRD, and Kucab resources for making reusable cancer-genomics datasets available to the community.

References

1. Alexandrov LB, Kim J, Haradhvala NJ, et al. The repertoire of mutational signatures in human cancer. *Nature*. 2020;578:94-101. doi:10.1038/s41586-020-1943-3.
2. Kucab JE, Zou X, Morganella S, et al. A compendium of mutational signatures of environmental agents. *Cell*. 2019;177:821-836.e16. doi:10.1016/j.cell.2019.03.001.
3. Ellrott K, Bailey MH, Saksena G, et al. Scalable open science approach for mutation calling of tumor exomes using multiple genomic pipelines. *Cell Systems*. 2018;6:271-281.e7. doi:10.1016/j.cels.2018.03.002.
4. Liu J, Lichtenberg T, Hoadley KA, et al. An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell*. 2018;173:400-416.e11. doi:10.1016/j.cell.2018.02.052.
5. Knijnenburg TA, Wang L, Zimmermann MT, et al. Genomic and molecular landscape of DNA damage repair deficiency across The Cancer Genome Atlas. *Cell Reports*. 2018;23:239-254.e6. doi:10.1016/j.celrep.2018.03.076.
6. Davies H, Glodzik D, Morganella S, et al. HRDetect is a predictor of BRCA1 and BRCA2 deficiency based on mutational signatures. *Nature Medicine*. 2017;23:517-525. doi:10.1038/nm.4292.

7. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 2016:785-794. doi:10.1145/2939672.2939785.
8. Cox DR. Regression models and life-tables. Journal of the Royal Statistical Society: Series B. 1972;34:187-220.
9. Harrell FE Jr, Lee KL, Mark DB. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. Statistics in Medicine. 1996;15:361-387. doi:10.1002/(SICI)1097-0258(19960229)15:4<361::AID-SIM168>3.0.CO;2-4.
10. Sanjaya P, Maljanen K, Katainen R, et al. Mutation-Attention (MuAt): deep representation learning of somatic mutations for tumour typing and subtyping. Genome Medicine. 2023;15:47. doi:10.1186/s13073-023-01204-4.

Supplementary Material

Supplementary Table S1. Supplementary endpoint inventory.

Endpoint	N	Task	Source	Metric(s)	Representation families
BRCA-pathway mutated	10,224	Binary classification	Regenerated benchmark output	AUROC	COSMIC NNLS exposures; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
DFI survival	5,147	Survival	TCGA MC3 clinical endpoint panel	Harrell C-index	Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack
DSS survival	9,530	Survival	TCGA MC3 clinical endpoint panel	Harrell C-index	Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack
Ever smoker	2,798	Binary classification	TCGA MC3 clinical endpoint panel	AUROC; Balanced accuracy	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
HRD LOH	772	Regression	TCGA-BRCA HRD labels with MC3 mutation features	Spearman r	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
HRD LST	772	Regression	TCGA-BRCA HRD labels with MC3 mutation features	Spearman r	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
HRD TAI	772	Regression	TCGA-BRCA HRD labels with MC3 mutation features	Spearman r	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
HRD24 high/low	772	Binary classification; Classification	TCGA-BRCA HRD labels with MC3 mutation features	AUROC	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
HRD42 high/low	772	Binary classification; Classification	TCGA-BRCA HRD labels with MC3 mutation features	AUROC	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
High stage	6,404	Binary classification	TCGA MC3 clinical endpoint panel	AUROC; Balanced accuracy	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
High tumor purity	9,591	Binary classification	TCGA MC3 clinical endpoint panel	AUROC; Balanced accuracy	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
LUAD KMT2C mutated	515	Binary classification	TCGA MC3 LUAD driver endpoint	AUROC	Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
Low-burden Kucab	259	Multiclass classification	Kucab mutagen-treated clone mutation catalogues	Balanced accuracy	Mechanistic controls; Mutational signatures
MMR-pathway mutated	10,224	Binary classification	Regenerated benchmark output	AUROC	COSMIC NNLS exposures; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
Original Kucab	259	Multiclass classification	Kucab mutagen-treated clone mutation catalogues	Balanced accuracy	Mechanistic controls; Mutational signatures
PARPi7	772	Regression	TCGA-BRCA HRD labels with MC3 mutation features	Spearman r	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
PARPi7 binary	772	Binary classification; Classification	TCGA-BRCA HRD labels with MC3 mutation features	AUROC	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
PFI survival	9,857	Survival	TCGA MC3 clinical endpoint panel	Harrell C-index	Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack
POLE/POLD1 mutated	10,224	Binary classification	Regenerated benchmark output	AUROC	COSMIC NNLS exposures; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry
TCGA 20 Type eCARD	60	Multiclass classification	Regenerated benchmark output	Accuracy	MuAt-compatible
	772	Regression	TCGA-BRCA HRD labels with MC3 mutation features	Spearman r	COSMIC NNLS exposures; Channel KME; Event-level MAF stack; Mutational burden; Mutational signatures; Signatures + MAF stack; UGA geometry

Note. Endpoint-level inventory for supplementary analyses outside the five-endpoint main panel.

Supplementary Table S2. Headline supplementary results.

Endpoint	N	Metric(s)	Best baseline/event-level	Best geometry/sensitivity	Best overall	Interpretation
BRCA-pathway mutated	10,224	AUROC	Signatures + MAF stack / XGBoost (AUROC 0.995)	COSMIC NNLS exposures / Source model (AUROC 0.820)	Signatures + MAF stack / XGBoost (AUROC 0.995)	Best baseline/event-level result exceeds sensitivity result by 0.175.
DFI survival	5,147	Harrell C-index	Mutational signatures / Cox PH (Harrell C-index 0.592)		Mutational signatures / Cox PH (Harrell C-index 0.592)	Best result comes from baseline or event-level feature families.
DSS survival	9,530	Harrell C-index	Mutational signatures / Cox PH (Harrell C-index 0.591)		Mutational signatures / Cox PH (Harrell C-index 0.591)	Best result comes from baseline or event-level feature families.
Ever smoker	2,798	AUROC; Balanced accuracy	Signatures + MAF stack / XGBoost (AUROC 0.781)	COSMIC NNLS exposures / Source model (AUROC 0.782)	COSMIC NNLS exposures / Source model (AUROC 0.782)	Best sensitivity result exceeds baseline by 0.001.
HRD LOH	772	Spearman r	Signatures + MAF stack / XGBoost (Spearman r 0.674)	UGA geometry / Source model (Spearman r 0.601)	Signatures + MAF stack / XGBoost (Spearman r 0.674)	Best baseline/event-level result exceeds sensitivity result by 0.073.
HRD LST	772	Spearman r	Mutational signatures / Source model (Spearman r 0.737)	UGA geometry / Source model (Spearman r 0.749)	UGA geometry / Source model (Spearman r 0.749)	Best sensitivity result exceeds baseline by 0.013.
HRD TAI	772	Spearman r	Signatures + MAF stack / XGBoost (Spearman r 0.762)	UGA geometry / Source model (Spearman r 0.721)	Signatures + MAF stack / XGBoost (Spearman r 0.762)	Best baseline/event-level result exceeds sensitivity result by 0.041.
HRD24 high/low	772	AUROC	Mutational signatures / Source model (AUROC 0.883)	UGA geometry / Source model (AUROC 0.898)	UGA geometry / Source model (AUROC 0.898)	Best sensitivity result exceeds baseline by 0.015.
HRD42 high/low	772	AUROC	Mutational signatures / Source model (AUROC 0.963)	UGA geometry / Source model (AUROC 0.933)	Mutational signatures / Source model (AUROC 0.963)	Best baseline/event-level result exceeds sensitivity result by 0.030.
High stage	6,404	AUROC; Balanced accuracy	Signatures + MAF stack / XGBoost (AUROC 0.639)	COSMIC NNLS exposures / Source model (AUROC 0.632)	Signatures + MAF stack / XGBoost (AUROC 0.639)	Best baseline/event-level result exceeds sensitivity result by 0.008.
High tumor purity	9,591	AUROC; Balanced accuracy	Signatures + MAF stack / XGBoost (AUROC 0.923)	UGA geometry / Source model (AUROC 0.638)	Signatures + MAF stack / XGBoost (AUROC 0.923)	Best baseline/event-level result exceeds sensitivity result by 0.286.
LUAD KMT2C mutated	515	AUROC	Signatures + MAF stack / XGBoost (AUROC 0.747)	UGA geometry / Elastic net (AUROC 0.746)	Signatures + MAF stack / XGBoost (AUROC 0.747)	Best baseline/event-level result exceeds sensitivity result by 0.001.
Low-burden Kucab	259	Balanced accuracy	Mutational signatures / Source model (Balanced accuracy 0.225)	Mechanistic controls / Source model (Balanced accuracy 0.274)	Mechanistic controls / Source model (Balanced accuracy 0.274)	Best sensitivity result exceeds baseline by 0.049.
MMR-pathway mutated	10,224	AUROC	Signatures + MAF stack / XGBoost (AUROC 0.999)	COSMIC NNLS exposures / Source model (AUROC 0.862)	Signatures + MAF stack / XGBoost (AUROC 0.999)	Best baseline/event-level result exceeds sensitivity result by 0.136.
Original Kucab	259	Balanced accuracy	Mutational signatures / Source model (Balanced accuracy 0.274)	Mechanistic controls / Source model (Balanced accuracy 0.357)	Mechanistic controls / Source model (Balanced accuracy 0.357)	Best sensitivity result exceeds baseline by 0.083.
PARPi7	772	Spearman r	Signatures + MAF stack / XGBoost (Spearman r 0.334)	UGA geometry / Source model (Spearman r 0.286)	Signatures + MAF stack / XGBoost (Spearman r 0.334)	Best baseline/event-level result exceeds sensitivity result by 0.048.
PARPi7 binary	772	AUROC	Mutational burden / Source model (AUROC 0.754)	UGA geometry / Source model (AUROC 0.755)	UGA geometry / Source model (AUROC 0.755)	Best sensitivity result exceeds baseline by 0.002.
PFI survival	9,857	Harrell C-index	Mutational signatures / Cox PH (Harrell C-index 0.568)		Mutational signatures / Cox PH (Harrell C-index 0.568)	Best result comes from baseline or event-level feature families.
POLE/POLD1 mutated	10,224	AUROC	Event-level MAF stack / XGBoost (AUROC 0.997)	COSMIC NNLS exposures / Source model (AUROC 0.854)	Event-level MAF stack / XGBoost (AUROC 0.997)	Best baseline/event-level result exceeds sensitivity result by 0.143.
TCGA 20 Type	60	Accuracy			MuAt-compatible / MuAt-compatible (Accuracy 0.050)	No compact comparison available; see technical table.
eCARD	772	Spearman r	Signatures + MAF stack / XGBoost (Spearman r 0.542)	UGA geometry / Source model (Spearman r 0.415)	Signatures + MAF stack / XGBoost (Spearman r 0.542)	Best baseline/event-level result exceeds sensitivity result by 0.127.

Note. Best baseline/event-level, geometry/sensitivity, and overall result for each supplementary endpoint.

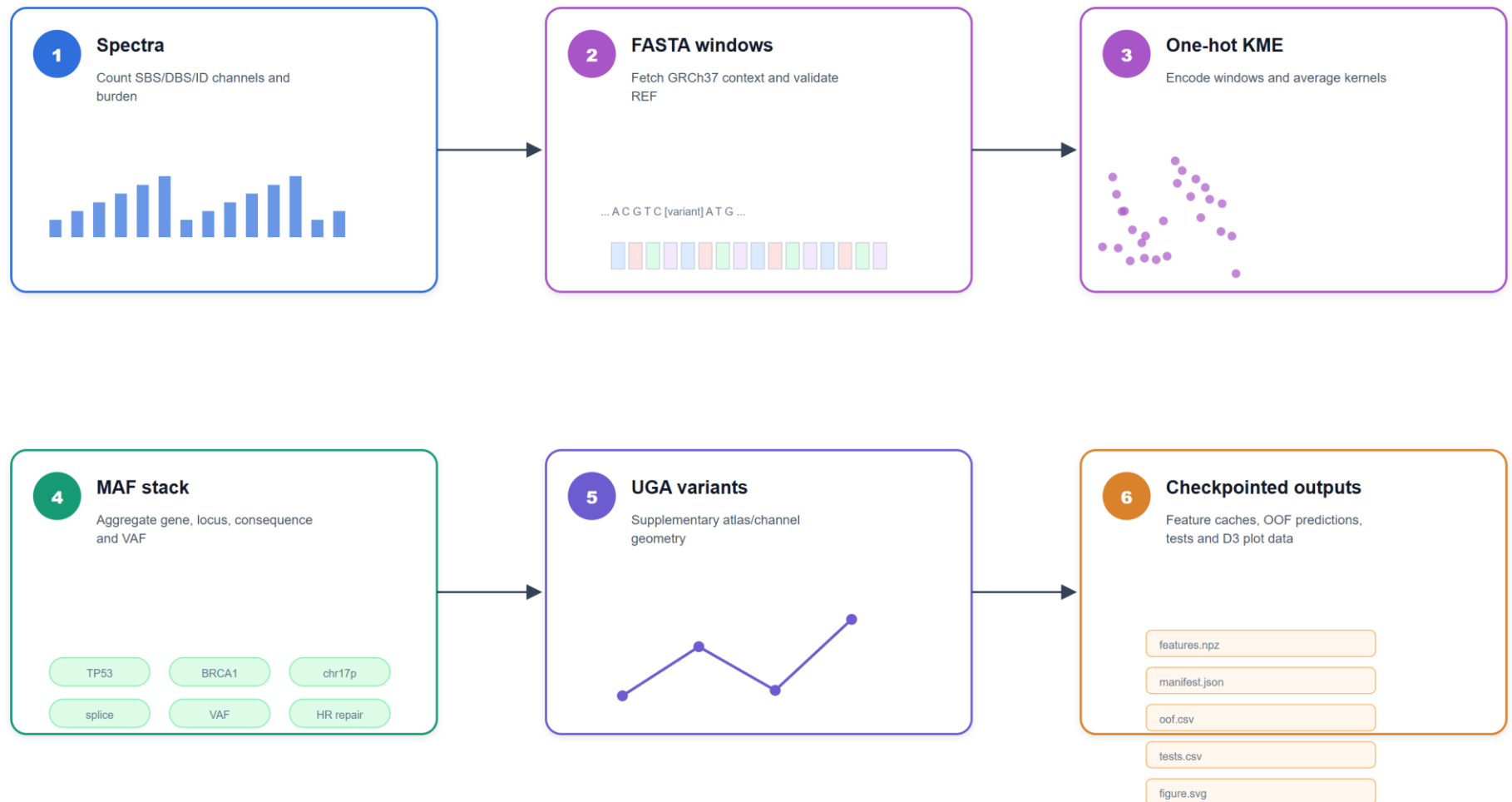
Supplementary Table S3. Non-applicability summary.

Representation/analysis family	Reason	Missing combinations	Example endpoints
COSMIC NNLS exposures	no measured supplementary result for this analysis family/endpoint	8	DFI survival; DSS survival; Low-burden Kucab; Original Kucab; PFI survival
Channel KME	no measured supplementary result for this analysis family/endpoint	10	DFI survival; DSS survival; Low-burden Kucab; Original Kucab; PFI survival
Mechanistic controls	no measured supplementary result for this analysis family/endpoint	20	DFI survival; DSS survival; HRD LOH; HRD LST; HRD TAI
UGA geometry	no measured supplementary result for this analysis family/endpoint	6	DFI survival; DSS survival; Low-burden Kucab; Original Kucab; PFI survival

Note. Grouped explanation of unsupported, intentionally omitted, or not-applicable supplementary combinations.

Supplementary Figure S1. Representation construction

Each expensive feature block is cached with input fingerprints, FASTA provenance, parameters and sample manifests.

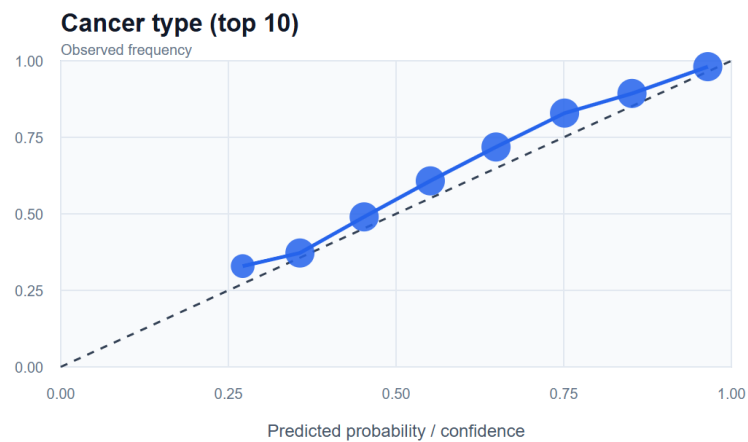
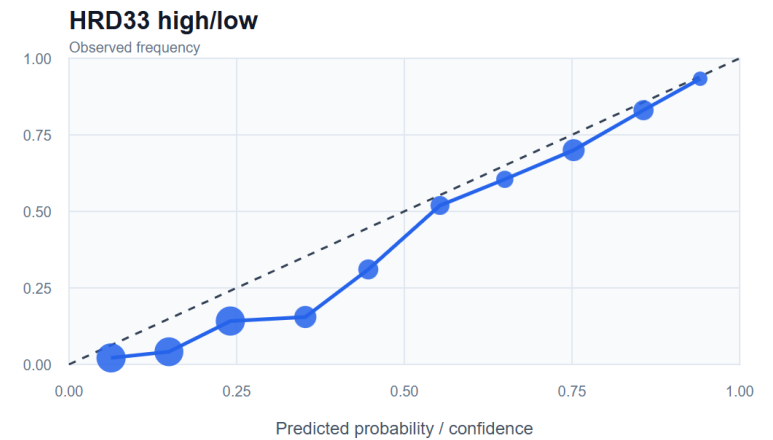
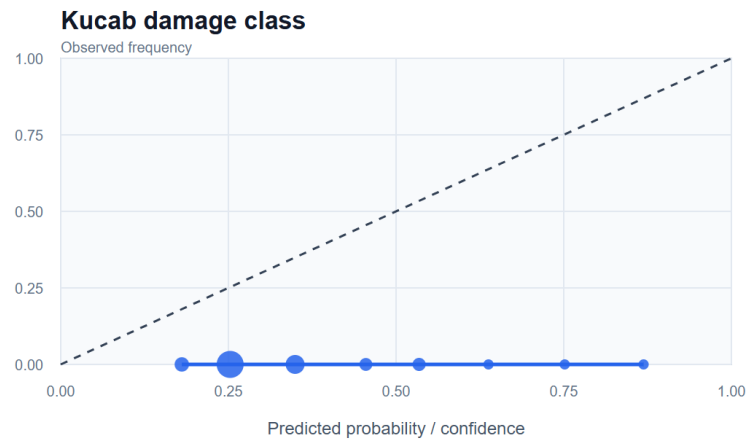


Supplementary Figure S1. Representation construction and reproducibility workflow.

Raw mutation catalogues are converted into spectra, geometry variants, MAF-stack aggregates, combined tabular matrices, and MuAt-compatible event bags with cached, restartable feature-generation steps.

Supplementary Figure S2. Calibration and reliability

Reliability curves from canonical OOF predictions for Signatures + MAF stack, XGBoost. Multiclass panels use confidence vs correctness.



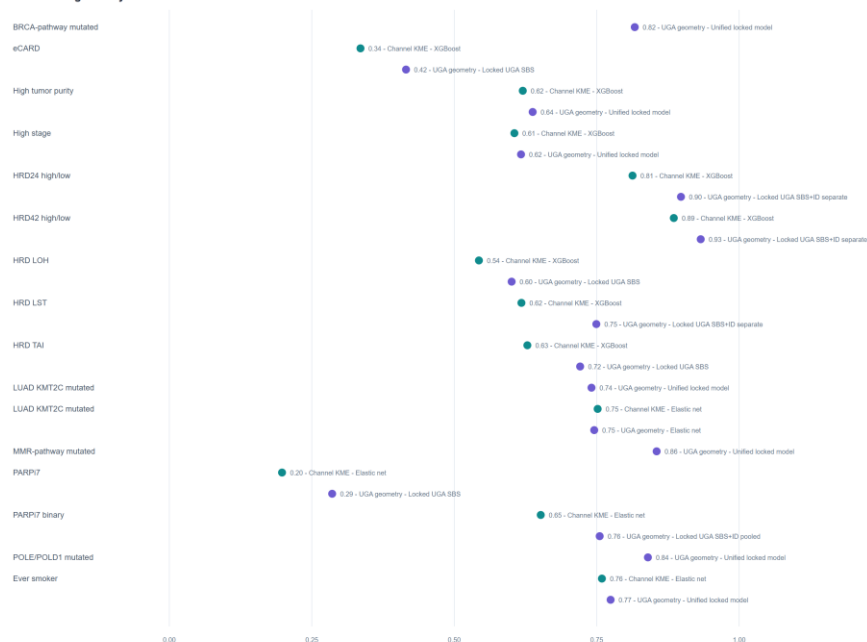
Supplementary Figure S2. Calibration of selected classification models.

Reliability curves are shown for selected out-of-fold classification models to assess whether predicted probabilities are broadly aligned with observed frequencies.

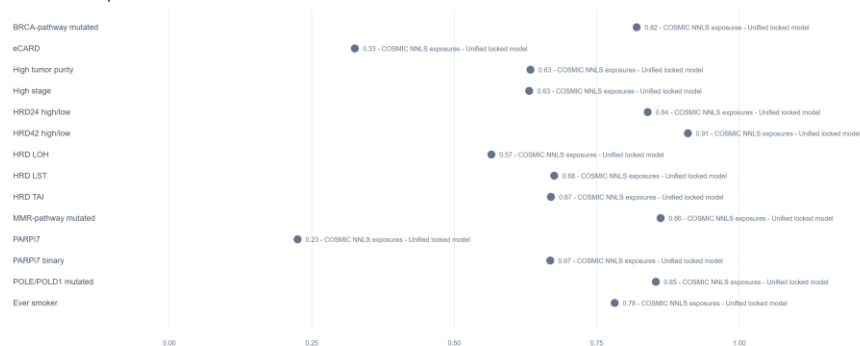
Supplementary Figure S3. Supplementary representation checks

Measured supplementary results only. Unsupported combinations are documented in Table S3 rather than drawn as placeholders.

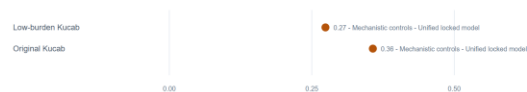
Alternative geometry



COSMIC/NLBS exposure checks



Mechanistic controls



Supplementary Figure S3. Supplementary measured representation panels.

Measured supplementary geometry, exposure, and mechanistic-control results are shown; unsupported combinations are omitted and documented in Supplementary Table S3.